Prime No.

1. Multiple of all primes up to prime $n+1$ = next prime.

2. All integers can be expressed as $(6k+i)$ for some integer k and $i = -1, 0, 1, 2, 3, 4$

   - All primes are of the forms $6k \pm 1$
     - Some Non primes can be also expressed in this form.

③ If any No. of the form $6k+1 \leq \sqrt{N}$ divides N then N is not a prime, else it is prime.

④ A simple but very inefficient primality test (Wilson Theorem)

$$(p-1)! = -1 \mod p$$

⑤ Let N be an odd positive integer. Let $N-1 = 2^k m$ for some $k > 1$ and some odd integer m. If for all integers n between 1 and $2(\log n)^2$ one has

$n^m \equiv 1 \pmod N$ or $n^{2^i m} \equiv -1 \mod N$ for some $0 < i < k$. Then N is a prime No.

Probabilistic Test (Fermat primality Test)

- Given an integer N, choose some integer a coprime to N and calculate $a^{N-1} \mod N$
  if $a^{N-1} \mod N \neq 1$ then N is composite
  else N may or may not be prime.
- But usually N is prime if $a^{N-1} \mod N = 1$

If $N = 341$ and $a = 2$ then

$2^{340} \equiv 1 \mod 341$

even though $341 = 11 \cdot 31$ is composite.

341 is smallest pseudoprime base 2

There are only 21853 pseudoprime base 2 that are less than $2.5 \cdot 10^9$

- This means, for N up to $2.5 \cdot 10^9$, If $2^{N-1} \mod N = 1$ then N is prime, unless N is one of these 21853 pseudoprimes.

- It is often used if a rapid screening of numbers is needed, for instance in the key generation phase of RSA.

Fast modulo operation

$x \bmod 2^N == x \,\&\, (2^N - 1)$ where $\&$ is bitwise AND.

It is better to choose $P = 2^K + 1$ for RSA.

$PK_1 + QK_2$

if $K_1 > K_2$ and $K_1 - K_2 < \delta$ then it is better evaluate in this fashion

$$PK_1 + QK_2 = P(K_1 - K_2) + K_2(P + Q)$$

NAF -

Digital signature - (Due to invasion purpose, use Ecc)

RSA & Extended Eclidian Algo.